INSTRUCTIONS

GEK-36456A

OBSOLETE

SINGLE SHEET

CLASSIFIER

3S7517SC250

GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

SINGLE SHEET CLASSIFIER

3S7517SC250

INTRODUCTION

These instructions describe the General Electric single sheet classifying equipment, 3S7517SC250, its installation, operation and maintenance. It is to be used as a guide by competent electronic service personnel.

Today's shear lines not only cut strip material into sheets with high accuracy, but also include equipment which inspects the material for various types of defects. In addition, there are provisions for the separate piling of prime and defective sheets. The control functions, from the receipt of a defect signal to the operation of the reject gates, are performed by the single sheet classifier.

This equipment accepts defect signals from the various strip inspection stations (pin hole detector, thickness gage, surface inspector, etc.) on continuously operated shear lines. It then delays these signals, in accordance with the strip speed, until the defect passes the shear cut line. After the cut, the defect signals are related to an individual sheet and again delayed in accordance with speed. Finally, when the sheet containing the defect nears the proper reject gate, the gate is operated and the defective sheet is diverted into one of the two reject piles. The equipment also includes set up switches which are used to select the final destination of defective sheets.

The single sheet classifier utilizes C-MOS (Complementary-Metal Oxide Semiconductor) integrated logic circuits on plug-in circuit boards. It also contains a built-in test circuit which permits a continuous survey of operation as well as faster servicing and maintenance.

RECEIVING, HANDLING AND STORAGE

Immediately upon receipt, the equipment should be examined for any damage which may have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company, and the nearest General Electric Sales Office should be notified promptly.

When the equipment is received, it should be unpacked carefully to avoid damage. Particular care should be exercised to prevent small parts from being mislaid or thrown away with the packing material.

If the equipment will not be used as soon as it is unpacked, it should be stored in a clean, dry place where it is protected from accidental damage. The equipment should not be stored in a location where construction work is in progress.

DESCRIPTION

The sheet classifying equipment consists of:

1. A control console, 3S7517SC250, which contains: the input isolators, logic circuits, test circuits, logic power supplies, and the gate switching circuits. The console is floor mounted and is to be attached to the shear line operator's console. The test panel is located in a pinnacle on the console. The front portion of the console is empty, providing space for other line control or monitor equipment which is normally supplied by others.

2. The "fine" pulse tachometer, 3S7517PG250A1, which produces 22 pulses per revolution. This unit consists of a housing and a slotted disc assembly and when properly geared to the shear feed mechanism, it produces pulse signals which accurately correspond to incremental strip movements. Also included with this unit is a 933B195G2 lamp supply transformer.

3. The "conveyor" pulse tachometer, 3S7517PG250B1. This unit is similar to the "fine" pulse tachometer except for its disc assembly which has only 2 slots. When properly geared to the conveyor, this unit produces pulse signals which correspond to incremental sheet movements. It, too, is supplied with a 933B195G2 lamp transformer.

4. Three sheet edge detectors (supplied by others).

5. Miscellaneous control contacts (supplied by others).

INSTALLATION

LOCATION AND MOUNTING

The console is similar in design to the main operator's console, and provisions are made to attach the two units together. The consoles are normally positioned so that the operator faces the line, the feed end on his right. When laying out the equipment, refer to the equipment outlines and the customer connection diagrams. Be sure to provide sufficient space to allow access to all units.

All of the equipment is designed for indoor use in an atmosphere free of corrosive and explosive fumes and as free of moisture as is possible. The ambient temperature should not exceed 40°C (104°F). If this temperature is exceeded, it may be necessary to install ventilating or cooling equipment.

All equipment should be rigidly mounted on a vibration free base.

ELECTRICAL CONNECTIONS

All electrical connections must be made in accordance with the customer connection diagram supplied with the equipment. Also, the customer must see to it that interconnections meet all applicable wiring codes.

All signal leads must be run in conduits which are separate from all other wiring - particularly power wiring.

Logic circuitry operating at low signal levels is sensitive to transient voltage spikes. Particular care was taken in this design to avoid this problem; however, in addition to careful routing of external signal leads, proper grounding is also of extreme importance; therefore, a good connection from the control cabinet to a stable earth ground is essential. In addition, all machine parts on the shear line should also be connected to the same electrical ground. Note that grounding for transient suppression requires much heavier conductors than used in normal grounding practice. Heavy copper braid is superior to normal solid conductors for this purpose.

PRINCIPLES OF OPERATION

GENERAL

The first sheet of the elementary diagram supplied with the equipment shows a functional block diagram of the classifier system. The upper part of this drawing shows the relative positions of the inspection stations, the various control devices and the reject gates.

The signals from the inspection devices are fed into isolator circuits, or, in the case of certain contact inputs, to relays. The purpose of the isolator circuit (or relay) is three fold; first, as its name suggests, it electrically isolates the classifier circuitry from the input device. Second, it acts as a noise filter. Finally, it is the means for converting the various types of input signals (contact or voltage pulses) into logic signals which are compatible with the internal circuitry. The output of the isolators is directly convertible to a logic signal while the relay contacts are fed into a latch circuit which produces the actual logic signal.

All of the input signals are routed through a "run-test" and test input gating arrangement to an appropriate shift register input or control element.

Each of the shift register blocks (32SR) contains 32 memory bits. An input signal, once set into a particular bit in the series string of 32 bits, is stepped or shifted along the string. This is accomplished by applying a shifting pulse (from the pulse tachometer) to all of the bits in the string. When this pulse occurs, any signal stored in the register is shifted forward by one bit.

It follows, then, that the number and time between shift pulses in conjunction with the number of storage bits will determine how long it takes for an input signal to appear at the output of the shift register. Because the shift pulses are produced by a pulse tachometer which provides one pulse for a predetermined strip (or sheet) motion, it can be seen that a defect signal in the register will move through the register in a manner simulating the actual strip motion from the inspection station toward the shear. It also follows that, if the proper number of bits are included in the register, the actual defect will arrive at the shear at the same time the defect signal reaches the end of the register.

The shift registers are composed of "D" type flip-flops (refer to logic element description for operation). In the register, each flip-flop applies its output signal to the input of the next. When a shift pulse is applied to all flip-flops, each assumes the state of the one preceding it. A signal is introduced into the register by directly setting one or more of the flip-flops. This is accomplished with adjustable

jumpering screws. A zero volt signal (logic 0) on this input causes no action, while a +5V signal (logic 1) represents a defect. For this reason, all unused jumper screws must be connected to zero volts. Memory retention of these circuits, even with power shutdowns, is assured by the use of extremely low power, complementary-MOS (C-MOS) integrated logic circuits and a standby battery. Functional details regarding the integrated circuits are covered in the "Description of Logic Elements and Uses" section of this book.

The shear cut line is representative of a changeover in information processing methods. At the time of cut, the classifier need no longer closely follow individual defects. At this point, it only needs to know whether or not there is a defect in the sheet which is being cut. Therefore, at the end of each strip register, there is an accumulator storage flip-flop. These flip-flops set when a defect signal reaches the end of the appropriate register before the cut. In this manner, they indicate that the sheet, which is about to be cut, contains a certain type of defect (pin hole, visual or gage). The accumulators set on the first signal they receive; therefore, they only indicate the type of defect - not the number of defects.

If a sheet contains more than one type of defect, an override function is activated. This circuit establishes a priority schedule and the overridden signals are cancelled accordingly (the accumulator flip-flop is reset). The override function is detailed in the adjustments section.

At the moment the shear cuts, it produces a contact closure (shear shift) which is fed through the contact isolator and the "run-test" circuits to a one shot multivibrator circuit. This circuit produces the shear shift signal (narrow pulse - approximately 1 ms) which is fed to the accumulator flip-flops and gating circuits. It is this signal which gates the accumulator information, pertaining to the sheet which was just cut, into the "open 1" or "input conv. reg." flip-flops. The trailing edge of the shear shift signal resets all of the register accumulators, readying them for the next block of sheet information.

If it is assumed that defect information was transferred to the "Open 1" flip-flop, the classifier now knows that a defective sheet is on the conveyor and its destination is reject gate #1.

The defect information will remain in the "open 1" flip-flop until the leading edge of the defective sheet activates the "open 1" proximity sensor. When this occurs, the proximity detector produces a signal which is fed through an isolator and the "run test" circuits to the "open 1" and "hold" accumulators. It is also fed to a one shot multivibrator. The leading edge of this signal transfers the information from the "open 1" flip-flop to the "hold" flip-flop and resets the "open 1" flip-flop (this action would reset the "hold" flip-flop if it were set and the "open 1" is reset).

When the "hold accum" sets, the classifier knows that the defective sheet is nearing gate #1. The gate could be opened at this time if it were clear. This condition is determined by the "hold" proximity sensor and a first sheet detector circuit. If the "hold" sensor is clear (no sheet under it), the first sheet detector circuit produces a signal (derived from the "open 1" sensor signal) which occurs approximately 0.1 ms after the "open 1" shift signal. This signal transfers the defect information from the "hold accum." flip-flop to the "gate 1 on" flip-flop, whose output controls the static switch powering gate #1. On the other hand, if there is a preceding sheet at the gate, no action will take place until its trailing edge clears the "hold" sensor. At this time, a signal is produced which transfers the sheet data from the "hold accum" flip-flop to the "gate 1 on" flip-flop.

Once set, the "gate 1 on" flip-flop (and gate 1 through a static switch) will remain on until the next good sheet (or bad sheet headed for gate #2) completes the cycle. This following sheet will first reset the "hold accum" flip-flop and then the "gate 1 on" flip-flop, which turns gate #1 off. This is a memory action which prevents gate operations between successive bad sheets.

If it is assumed that defect information is transferred into the "input conv reg" flip-flop, rather than the "open 1" flip-flop, a different set of actions takes place. The setting of the "input conv reg" flip-flop indicates to the classifier that there is a defective sheet on the conveyor whose destination is gate #2.

The defect information will remain in the "input conv reg" flip-flop until the leading edge to the defective sheet produces a signal from the "open 1" proximity sensor. The signal from the sensor is applied to a one shot multivibrator whose output signal is a narrow pulse. This pulse starts on the leading edge of the "open 1" sensor signal and lasts for approximately 0.1 ms. It gates the defect information into the conveyor shift register and then resets the "input conv reg" flip-flop. When this takes place, the classifier knows that there is a defective sheet at the "open 1" sensor and it is on its way to gate #2.

Next, the "hold" proximity sensor, in conjunction with the "first sheet" detector, produces appropriate signals which will close gate #1 - if it were open (refer to details in preceding description).

Upon entering the conveyor shift register, the defect signal is shifted through the register at a rate determined by the conveyor pulse tachometer. The circuits and operation of this register are similar to the strip registers. In this register, however, the only requirement for proper information flow is that the defect signal must arrive at the "gate 2 accum" after the leading edge of the preceding sheet actuates the "open #2" proximity sensor - but before the leading edge of the defective sheet arrives at the "open 2" proximity sensor. This permits a range of sheet lengths to be handled by the conveyor shift register; but, if sheet length is changed beyond this range, it will be necessary to readjust the register length.

When the defect signal arrives at the end of the conveyor shift register, it sets the "gate 2 accum" flip-flop. It will remain in this flip-flop until the leading edge of the defective sheet operates the "open 2" sensor. At this time, the "open 2" sensor produces a signal which is fed through the isolator and "run-test" circuits to the "gate 2 accum" and "gate #2 on" flip-flops. This signal will cause the "gate 2 on" flip-flop to set (gate #2 will energize through a static switch) and the "gate 2 accum" flip-flop to reset.

The "gate 2 on" flip-flop will remain set (gate #2 energized) until the leading edge of the next good sheet activates the "open 2" sensor. When this occurs, the "gate 2 on" flip-flop resets and gate #2 deenergizes. As at gate 1, this is a memory action which eliminates unnecessary gate operations between adjacent defective sheets.

MISCELLANEOUS FUNCTIONS

Provisions are made to accept a contact closure from a reject station (normally a pushbutton) located at the shear. The contact closure picks up relay SVR and applies a reject signal to the "visual accumulator" flip-flop.

Under special conditions, established by others, it is necessary to prevent normal reject gate operation. When this "lockout" function is required, an external contact closure causes relay LOK to pick up and a contact from this relay prevents normal operation of both gates.

Terminals are available which permit manual gate operation. A contact closure between the proper terminals will energize either gate 1 or gate 2. This function is primarily meant for test purposes - WARNING - IT SHOULD NEVER be used when the line is operating unless external synchronizing and interlocking circuits are employed.

A special circuit is included which energizes static switch MSS when there is a sheet with a pin hole about to be sheared. If this condition occurs, the static switch will energize when shear operated contact "A" closes. It deenergizes when shear operated contact "B" opens (refer to initial adjustments for contact set up). The output of this circuit can be used to initiate an external marker circuit.

POWER SUPPLIES

The classifier circuits and peripheral equipment are powered from a 115 VAC, 50/60 Hz. power line through a customer-supplied fused disconnect.

Within the classifier, on the main panel at the rear of the console, the 115V power is switched (LSW), fused and distributed. It is applied to outgoing terminals for the external equipment, through switches (1GSW and 2GSW) to the gate circuits and also through a line filter to the DC power supplies. The line filter along with a spike suppression device (1MOV, metal oxide varistor) reduce the effects of line voltage transients on circuit operation.

There are three DC voltage supplies on the main panel. First, a +5 volt isolator/relay supply. This supply is regulated and is used to power all isolators, relays, and the pulse tachs. Second, the 5.2 volt logic supply. This unit is identical to the isolator supply; however, its output is set to a slightly higher DC voltage (5.2V). This supply powers the internal logic circuits only. By using separate supplies in this manner, the possibility of coupling noise signals through the power supplies is greatly reduced. Finally, a standby battery supply is included. This is a sealed rechargeable, jelled- electrolyte lead-acid battery. It provides standby power to the logic circuits whenever the 115 VAC power is removed and thereby prevents the loss of information from the logic circuits. Because of the low power C-MOS logic circuits, this battery can maintain the "memory" for months; however, in normal usage, this function would be required only for overnight or weekend shutdowns. The battery is maintained by "floating" it

on the 5.2V logic supply. When used in this manner, the battery should provide many years of service. If for some reason a replacement battery of this type is not readily available, alternate provisions have been made to use three "D" size flashlight cells. Under normal use, these cells should last for several months (normally up to the shelf life of the battery). The condition of the batteries can be checked by simply measuring the voltage across them in the "Test" mode. If this voltage is below 4.0V, replace the batteries.

TEST AND CONTROL CIRCUITS

Test circuits and a test panel are provided on the console pinnacle. These can be used to simulate input signals and signal flow through the classifier circuits. Details regarding the various switch functions are included in the "initial adjustment" section. In addition, it is also important that certain of the circuit functions are understood. When the "run-test" switch is in "test" position, all external inputs are blocked; therefore, if a test for an external input is required, the "run-test" switch must be switched to "run". After verifying this operation, the classifier can then be switched to test, and this signal can be followed with the test circuits.

The test circuits are also interlocked with a "power up-down" circuit. This circuit is controlled by relay LVR which senses the application or loss of the AC line voltage. When line voltage is first applied, this circuit blocks all external inputs and the outputs for approximately 50 ms. This provides sufficient time for the DC supplies to build up and prevent the application of false input signals to the registers. With a loss of line voltage, the circuit rapidly blocks all inputs and outputs before the DC supplies fall. Again, this prevents false inputs. Note, however, that this circuit will not compensate for long warm-up times in external equipment; therefore, it is recommended that all external equipment be "powered up" before the classifier.

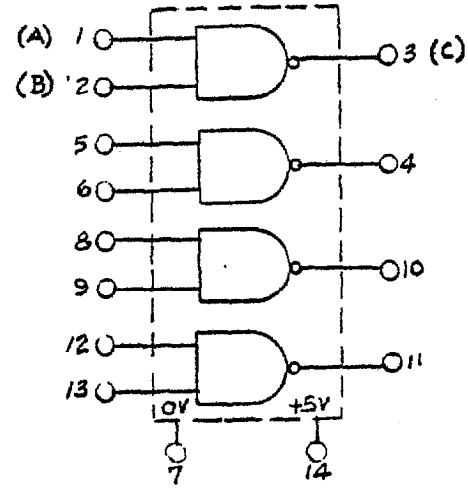
DESCRIPTION OF LOGIC ELEMENTS AND USES

This equipment uses standard C-MOS integrated logic circuits in 14 and 16 pin dual-in-line packages. In servicing the classifier, it is important that the functions and rules pertaining to the various logic elements are understood.

1. C-MOS Quad 2 Input NAND Gate, CD4011AE

Each 14 pin pack contains four 2-input NAND gates. The pin numbers, symbols and the NAND truth table are shown below.

PACK SYMBOL



Truth Table where A and B are Inputs and C is the Output.

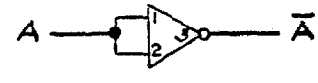
A	B	C
0V	0V	+5V
+5V	0V+	+5V
0V	+5V	+5V
+5V	+5V	0V

Either input = 0V
Then output = +5V

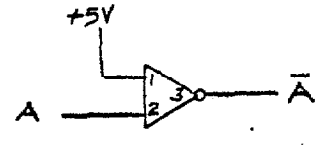
Both inputs = +5V
then output = 0V

Note that +5V = logic 1
and 0V = logic 0 (positive logic)

Symbol when used as an inverter.



OR



The 2-input NAND gate is one of the most basic logic elements. It is also one of the most versatile and, for this reason, it is used in many of the circuits in the classifier.

The basic NAND gate is used in the classifier to perform switching or blocking functions. Note that (refer to truth table) the output will not change if either input is held at 0V. But, the output will inversely follow one input if the other is held at +5V.

When properly connected, two NAND gates can be used as a basic latch (or flip-flop) circuit. The classifier ues this type of latch to transform the relay contact and test switch inputs into logic signals.

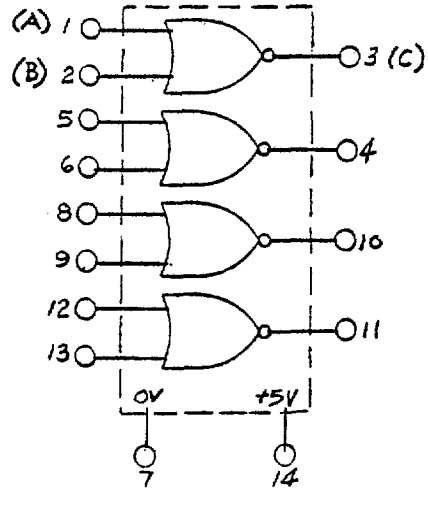
Two NAND gates can be simply connected to form a free-running pulse generator. This is accomplished by the use of an appropriate resistor-capacitor feedback network. In the classifier, this circuit is used to produce simulated shift pulses for use in the auto-shift test mode.

Two NAND gates can also be connected to form a simple one shot multivibrator. This circuit is used in the "shear" and "open 1" shift functions.

2. C-MOS Quad 2-Input NOR Gate, CD4001AE

Each 14 pin pack contains four 2-input NOR gates. The pin numbers, symbols and the NOR gate truth table are shown below.

PACK SYMBOL



Truth Table where A and B are inputs and C is the output.

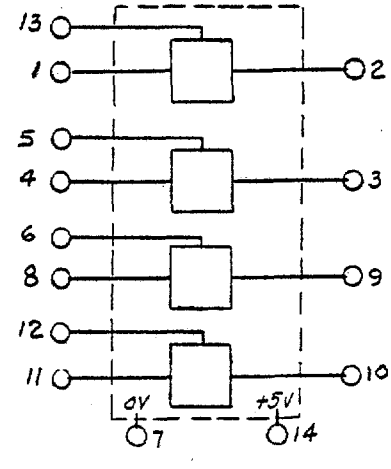
A	B	C	
0V	0V	+5V	Both inputs = 0V then output = +5V
0V	+5V	0V	Either input = +5V then output = 0V
+5V	0V	0V	
+5V	+5V	0V	

Note that +5V = logic 1
and 0V = logic 0

The 2-input NOR gate is also one of the most basic logic elements and it can be used in the same ways as the NAND gate. In the classifier, it is used to block the outputs to the indicator and gate drivers because (refer to truth table) when it is blocking, its output is zero volts.

3. C-MOS Quad Bilateral Switch, CD4016AE.

PACK SYMBOL (14 pins)

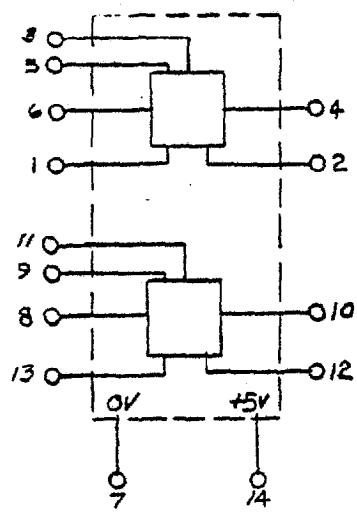


As its name suggests, this unit consists of 4 logic switches. It is used in the classifier in the override switching circuits. In operation, 0V on the control input (pin 13, for example) produces an open in the controlled circuit (between pins 1 and 2, for example); however, a +5V signal on the input (pin 13) will yield a low resistance path across the controlled circuit (between pins 1 and 2).

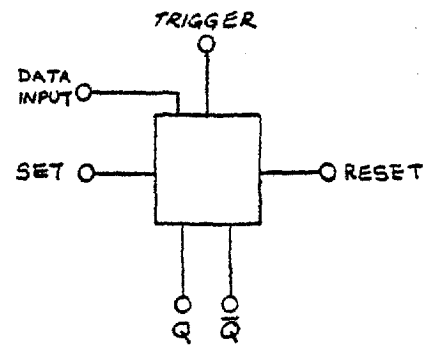
4. C-MOS Dual "D" Flip-Flop, CD4013AE

Each of these 14 pin packs contains two "D" type flip-flops. The symbol and truth table are shown below.

PACK SYMBOL



SIGNAL IDENTIFICATION (1/2)



Truth Table and Uses

Trigger	Data In	Reset	Set	Q	$\overline{Q}$	
⌐	0V	0V	0V	0V	1	
⌐	+5V	0V	0V	1	0	
⌙	X	0V	0V	Q	$\overline{Q}$	*
X	X	+5V	0V	0V	+5V	
X	X	0V	+5V	+5V	0V	

* No change

X = Either condition

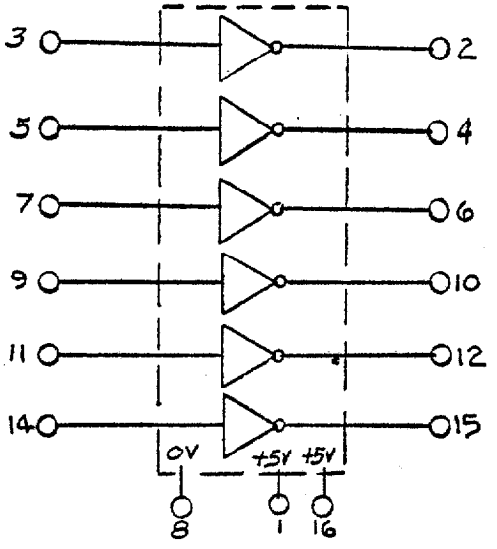
Simply stated, if a 0V signal is applied to the D input, the flip-flop will (if set) reset when the trigger input rises. If a +5V signal is applied to the D input, the flip-flop (if reset) will set when the trigger input rises. The fall of the trigger signal causes no action, while the set and reset inputs operate directly without regard for a trigger signal. These inputs will override a data signal.

The "D" flip-flop is used in the classifier to perform all of the memory functions. These include the shift registers, "shear" and "gate" accumulators and the gate memory. Note that these flip-flops, in conjunction with the battery supply, form a permanent or non-volatile memory in the system. This is made possible because of the extremely low power requirements of the C-MOS circuits.

5. C-MOS Hex Inverter, CD4009AE

Each 16 pin pack contains six inverters with a higher drive ability than most other C-MOS devices. This unit is used to interface between the ultra low power C-MOS circuit and higher power circuits or, connected in parallel, they can provide faster pulse rise times.

SYMBOL



Truth Table

Input	Output
0V	+5V
+5V	0V

INITIAL ADJUSTMENTS

1. Carefully wire check all connections before applying power.

2. CAUTION: DO NOT run the shear line until proper gate operation has been verified. Also (particularly with certain types of mechanical reject gates), it is advisable, for safety, to disconnect the gate at the control cabinet during the initial test and adjustment procedures.

3. Apply power to the classifier console by closing the main external customer-supplied disconnect switch.

4. Apply power to the logic circuits, pulse tachometers, and proximity switches by closing switch LSW (located on main panel - accessible when rear door of console is open). Note that 1GSW and 2GSW control the power to the gates - these two switches should be off for initial tests and adjustments.

5. Pulse tachometer adjustments (3S7517PG250A1 and B1).

a. Align the back plate and collar assembly on the shaft so that the shaft rotates freely without rubbing collar.

b. Position the rotating disc assembly per the dimension shown on the outline drawing.

c. Adjust the cell block so that the top of the small slot in the disc is centered in the inspection hole as shown in Figure 1A.

d. Adjust the lamp bracket so that each of the two photosensors receive approximately the same intensity of illumination. This may be accomplished by using the inspection hole to visually align the lamp filament.

NOTE: The lamp voltage should be decreased (this may be done by using a voltage dropping resistor in series with one lamp lead) to a point where the filament brilliance is reduced to a red glow. The filament should be centered in the one plane and below center in the other. See Figure 1B.

e. When adjustments are complete, tighten all mounting screws and recheck alignment.

6. Adjust the shear operated contacts; shear shift, A and B. Refer to Figure 2.

a. The shear shift contact should close for approximately 15° within the time of no feed and coinciding with the shear cut.

b. The B contact should close at the end of the feed cycle and remain closed for approximately 100° of the shear cycle. This contact must close before the A contact and remain closed until after the shear contact opens.

c. The A contact should close only after the B contact closes and remain closed for approximately 10°.

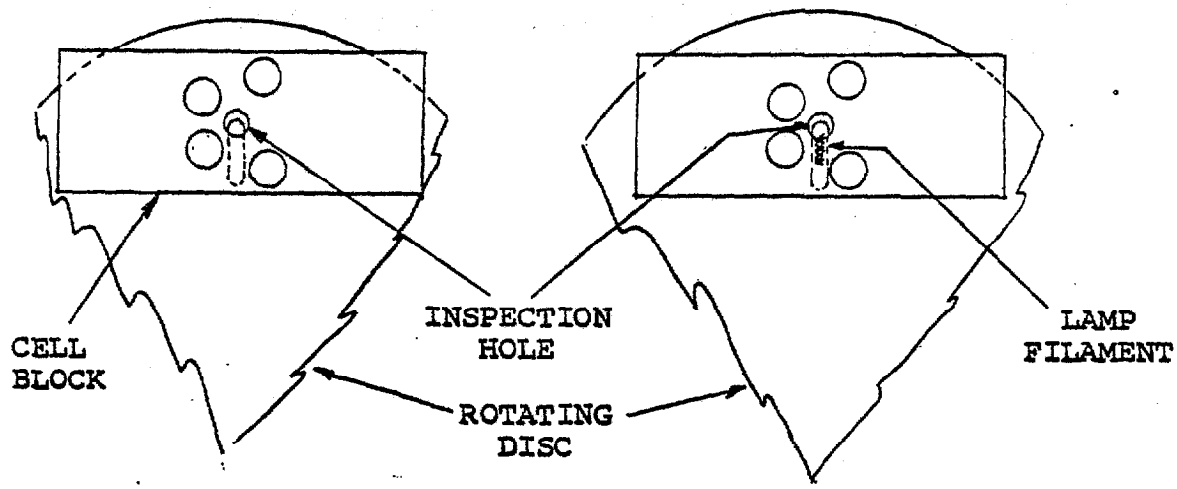


Figure 1A

Figure 1B

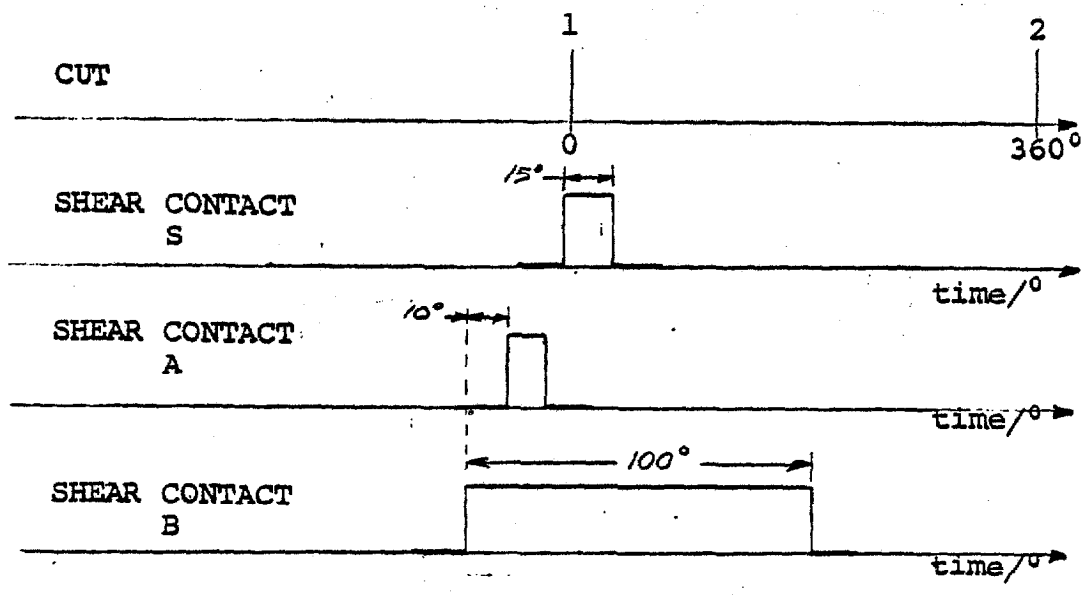


Figure 2

d. Secure adjustment screws and recheck all adjustments.

7. Adjust the shift register lengths to correspond to the shear line dimensions. Each of the shift registers (pin hole, visual; gage and conveyor) uses at least one 3S7517SR300 shift register board. This board can cause up to 32 delays. Each delay corresponds to an increment of strip (or sheet) travel as determined by the shear (or conveyor) pulse tachometer. There are two jumper adjustable inputs which can be preset to cause from zero to 31 delays on a particular board.

a. Adjust the pin hole register length:

NOTE: The pin hole register consists of one adjustable 32 bit register board (3S7517SR300) and a fixed 18 bit register located on the 3S7517TP250 board.

Procedure

1) Accurately measure the strip length from the pin hole detector center line to the shear "cut line" (where the shear cuts the sheet from the strip).

2) Find the total number of delay bits required in the register by dividing the length found in step 1) by the incremental distance the strip travels for each shear pulse tachometer signal (measure or calculate).

3) Round the number found in step 2) to the nearest integer.

4) Subtract 18 (number of fixed bits) from the number found in step 3). The difference resulting from this subtraction is the number of bits required on the 3S7517SR300 board in slot 4 (pin hole register board).

5) On the 3S7517SR300 board, from slot 4, move the jumpering screw (at the number found in step 4) from the center row (0 volts) to the outer row labeled input #1.

6) Setting the guard band: Note that because a defect can occur at any time between shift pulses, the highest possible accuracy for the transfer of information by the register is limited to one shift increment (usually about one inch). This means that if a particular defect were to be very close (within a shift increment) to the "cut line" at the time of shearing, the control may not be able to correctly identify the defective sheet. To eliminate this possibility, a so-called guard band is set up.

Instead of feeding the defect information into one bit, it is also fed into adjacent bits. Now, should a defect occur close to the "cut line", the defective sheet and the adjacent sheet will be rejected.

The guard band is set by moving the two (or more for wider bands) jumper screws which are adjacent to the one set in step 5), to the same outer row as used in step 5.

Example:

Set up the pin hole register when the strip length from pin hole detector to cut line (measured) is 46.4 inches and the shift increment is 1.1 inches (measured or calculated from pulse tachometer slots and the gearing ratios).

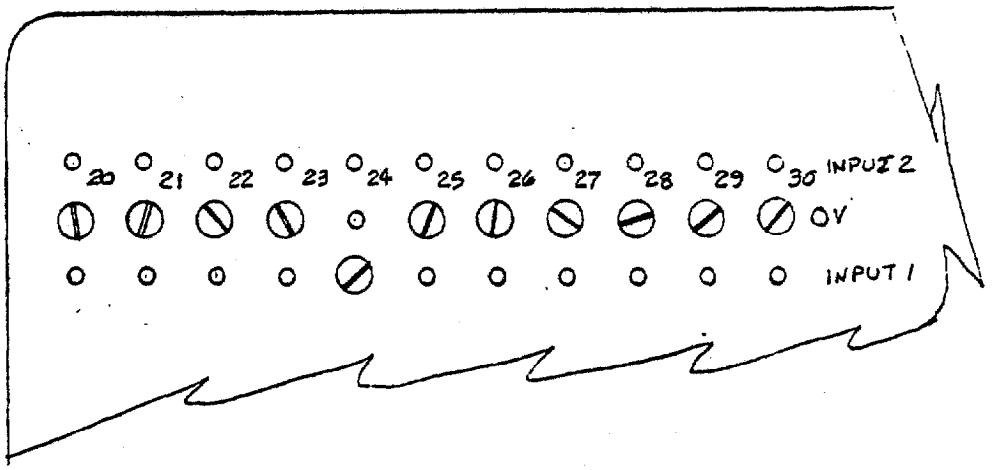
a) Measured strip distance = 46.4 inches.

b) Find total bits required 46.4/1.1 = 42.18.

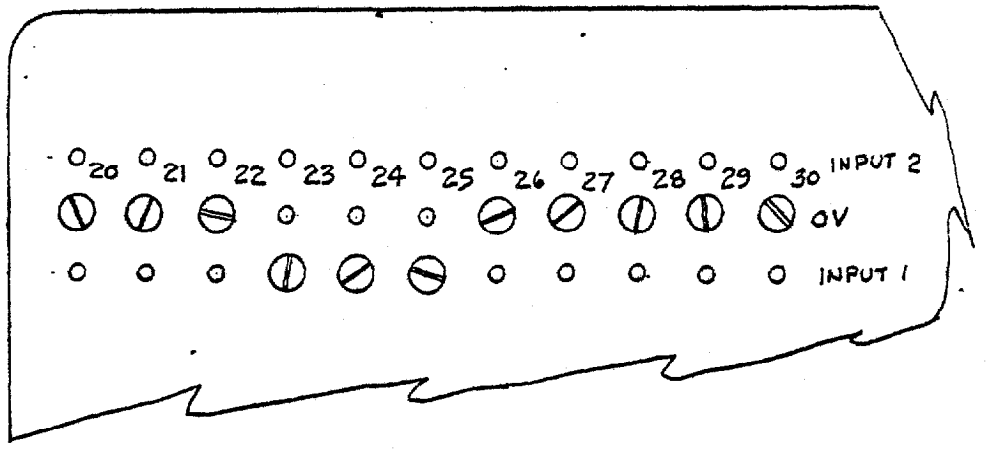
c) Rounding = 42.

d) Subtract 18: 42-18 = 24.

e) Move the jumpering screw labeled "24" from the center row (0 volts) to the appropriate outer row (input #1).



f) Set up the "guard band" by moving adjacent screws (23 and 25) to the same outer row.



g) The pin hole register is now set up properly for the conditions specified in the example.

b. Adjust the visual and gage register lengths: Note that the adjustment of these registers is similar to that used for the pin hole register, however, no fixed registers are employed and the shift increment is twelve times that used in the pin hole register.

1) Measure the strip length from the center line of the inspection station in question to the "cut line". Be sure to include the average loop length in this measurement.

2) Find the total number of delay bits required by dividing the length found in 1) by 12 times the incremental distance the strip travels for each shear pulse tachometer signal (found when setting the pin hole register).

3) Round the number found in step 2 to the nearest integer. This number equals the number of bits required on the 3S7517SR300 board in question (slot 2 for gage or slot 3 for visual).

4) Determine which input (1 or 2) is being used for the signal in question (see diagram - note that input term. B1 = input #1 and input term. A7 = input #2). On the 3S7517SR300 board, from the appropriate slot, move the jumpering screw (at the number found in step 3) from the center row (0 volts) to the appropriate outer row (1 or 2).

5) Set the guard band by moving the two screws, which are adjacent to the one set in step 4, to the same outer row as determined in step 4.

c. After the strip registers are set up, their accuracies should be verified. Mark the strip at the inspection station in question, generate a defect signal (either on the inspection device or by using the appropriate test input switch on the classifier test panel), then jog the shear line until the appropriate accumulator flip-flop sets (this is indicated on the classifier test panel). At that moment, the mark which was previously placed on the strip at the inspection station should be within one shift increment of the shear "cut line". If the ±1 shift increment guard band is used, the mark should be 1 shift increment before the "cut line". Make minor corrections in the jumper screws if required.

d. Setting the conveyor shift register: Note that the function of this register is similar to the strip registers; however, it follows a whole sheet rather than a small defect as in the pin hole register, for example. Also note that the information in this register need not be positioned with respect to a particular point on the defective sheet. The only requirement here is that the information in the conveyor register must arrive at the gate 2 accumulator flip-flop after the leading edge of the preceding sheet operates the open 2 proximity switch and before the leading edge of the defective sheet operates open 2.

Procedure.

1) Measure the distance from "open 1" to "open 2" proximity switch sensors.

2) Find the total number of delay bits required by dividing the distance found in 1) by the incremental distance a sheet travels on the conveyor for each conveyor pulse tachometer signal (measured or calculated).

3) Round the number (found in step 2) down to the nearest lower integer and subtract 2. This difference equals the number of bits required in the conveyor register.

4) On the 3S7517SR300 board from slot 1, move the jumpering screw (at the number found in step 3) from the center row (0 volts) to the outer row labeled input #1.

5) A guard band is not normally used in the conveyor shift register.

6) Verify by placing a sheet on the conveyor with its leading edge near the open 1 sensor, then insert a signal (using classifier test switches) at "input conv reg". Then jog the line until the "Gate 2 accum" flip-flop sets. Note the position of the sheet leading edge at this time and compare with requirements under d. Adjust if required.

8. Proximity sensor locations.

a. The "open 1" sensor is normally positioned just beyond the leading edge of the longest sheet which is to be cut. This setting will permit a range of shorter sheets to be cut, see "Operation" for restrictions. If the position of the "open 1" sensor is changed, it may be necessary to readjust the conveyor shift register.

b. The "Hold" sensor should be mounted as close as is practical to the #1 gate edge nearest the shear.

c. The "open 2" sensor should be located near the #2 gate edge nearest the shear, normally 8 to 10 inches.

9. Set modes of operation using "gate 1", "gage 2", "Vis." and "gage" selector switches. These switches are used to program the routing of certain defect types to either of the reject piles. This table

shows the possibilities available. Note that when more than one type of defect is present in one sheet, the override function is activated. Refer to override selector settings.

TYPE OF DEFECT	SHIFTS THROUGH REGISTER	TO REJECT PILE
Pin hole	P. H.	1 (always)
Visual	Vis.	1 or 2 (select with "VIS" switch)
Off Gage 1	Select with "GAGE 1" Sw { Vis / Gage	1 or 2 (select with "VIS" Switch) / 1 or 2 (select with GAGE Sw.)
Off Gage 2	Select with "GAGE 2" Sw { Vis / Gage	1 or 2 (select with "VIS" Sw.) / 1 or 2 (select with GAGE Sw.)

10. Use of the "override selector" switch.

The override function comes into play when a sheet contains more than one type of defect. When this occurs, a schedule of dominance or priority, preset by the override selector, is established. This action takes place at the three register accumulators (P, V, G). The following chart shows the priorities for the various "Override Selector" switch settings.

OVERRIDE SEL.	SWITCH POSITION	#1 PRIORITY	#2 PRIORITY	#3 PRIORITY
1	P/V/G	P	V	G
2	P/G/V	P	G	V
3	V/P/G	V	P	G
4	V/G/P	V	G	P
5	G/P/V	G	P	V
6	G/V/P	G	V	P

Once an override priority is established, the ultimate destination (gate 1 or gate 2) for a sheet with a particular defect or combination of defects is governed by the "VIS" and "gage" switches.

Any one of seven defect combinations can occur in a sheet. These are designated in the chart that follows as: 1. "P" (data from pin hole register); 2. "V" (data from visual register; NOTE, this register can contain gage defects (dependent upon the setting of the "G1" and "G2" switches at the input of the visual register) as well as visual; however, any sheet data received through the visual register will be designated "V" on the chart); 3. "G" (data from gage register); 4. "P" and "V"; 5. "P" and "G"; 6. "V" and "G"; 7. "P" and "V" and "G".

This chart can be used to determine what switch settings are required to separate, between the two reject piles, sheets with various defects or defect combinations. Note that there are duplications. Also note that there is a possibility of all defect combinations in gate 1 (with "Vis" switch = gate 1 and "Gage" switch = gate 1). This has not been

covered in the chart because the override function does not cause any change - all defective sheets regardless of combination will go into reject gate 1.

Example: Assume that it is desirable to stack all sheets containing pin holes in one pile and all others in the other.

a. Write the statement in a form which agrees with the chart - one pile should contain P+ PV + PG + PVG,, while the other should contain V + G + VG. Use the shorter expression (V+G+VG) in step b.

b. Search the A, B, and C sections, lines gate 1 and 2, of the chart until the expression V + G + VG is found. This is "C2" under column P/V/G and also C2 under P/G/V.

c. Set "Vis" and "Gage" switches as shown in C portion. Vis = gate 2, Gage = gate 2.

d. Set override selector to "P/V/G". Note that this expression is duplicated under the P/G/V setting. This means that both of these settings will result in the same separation.

e. Note that the gate at which the V + G + VG combination will be delivered is gate 2 while the P + PV + PG + PVG combinations will be at gate 1.

11. Classifier Test Panel

Once the principles of operation are understood, the switches and indicators on the test panel can be used to follow a signal through the classifier circuits. A brief description of the switch functions follows:

a. "Run-test" switch - when in "test", the external inputs are locked out. This switch must be in the "run" position for normal line operation.

b. The "PH Test", "VIS Test", "Gage 1 Test" and "Gage 2 Test" switches are used to simulate the respective defect signals.

c. The "Man Shift" produces a single simulated shear tachometer and "conv tach" pulse for each operation. Twelve operations are required to shift the visual and gage register data 1 bit.

d. "Run-Auto Shift" switch - when in "Auto shift" simulated shear tachometer and conveyor tachometer pulses are produced at approximately a 50 Hz rate. This switch must be in the "run" position for normal line operation.

e. The "Erase" switch resets all of the accumulator and gate memory flip-flops.

f. The "Shear shift", "Open 1 shift", "Hold shift" and "Open 2 shift" switches are used to simulate the respective shift signals.

12. Switch power "off" (LSW).

13. Make sure that the battery fuse (3FU), at location 6 on the main panel, is in the upper position. The lower position is for shipping or storage; it removes all load from the battery.

14. Reconnect the gate circuits.

15. Restore power to the classifier (LSW, "on") and apply power to the gate circuits (1GSW and 2GSW, on) with the line stopped. WARNING: Use extreme care because gate operation is now possible.

16. Check for proper operation of the gates using the test panel switches and then, if practical, manually with an actual sheet.

17. After verifying correct operation, check to see that the "run-test" switch and the "run-auto shift" switches are in the "run" position. The classifier should now be operational.

OPERATION

Once the line and classifying equipment has been adjusted, the classifier operation is fully automatic. Operator adjustments are required only if a different rejection program is desired. This is accomplished with the "Override Selector", "Gage 1", "Gage 2", "VIS", and "GAGE" switches. Refer to initial adjustments for details.

When sheet lengths are changed, it may be necessary (for extreme changes only) to reposition the "open 1" sensor. This sensor must always be located so that the leading edge of the sheet being sheared will energize the sensor after the shear cut (shear shift signal); therefore, it must be positioned at a distance from the shear which is slightly greater than the longest sheet to be cut.

If this is done for the longest sheet, the distance may be greater than two of the shortest sheets - if the sheet spacing is quite close. If the spacing is large enough, the free sheet will have already operated the

SECT.	GATE OR REJECT PILE	P/V/G	P/G/V	V/P/G	V/G/P	G/P/V	G/V/P	OVERRIDE SELECTOR POSITIONS ← / "VIS" AND "GAGE" SWITCH SETTINGS ↓
A	1	#1 duplicate P+V+PV+PG+PVG	P+V+PV+PG+PVG	#1 duplicate P+V+PV+PG+VG+PVG	P+V+PV+VG+PVG	#2 duplicate P+V+PV	#2 duplicate P+V+PV	VIS = gate 1
	2	G	G + VG	G	G + PG	G+PG+VG+PVG	G+PG+VG+PVG	gage = gate 2
B	1	P+G+PG+PV+PVG	#4 duplicate P+G+PG+PV+VG+PVG	#3 duplicate P+G+PG	#3 duplicate P+G+PG	#4 duplicate P+G+PG+PV+VG+PVG	P+G+PG+VG+PVG	VIS = gate 2
	2	V+VG	V	V+PV+VG+PVG	V+PV+VG+PVG	V	V+PV	gage = gate 1
C	1	#5 duplicate P+PV+PG+PVG	#5 duplicate P+PV+PG+PVG	P+PG	#6 duplicate P	P+PV	#6 duplicate P	VIS = gate 2
	2	V+G+VG	V+G+VG	V+G+VG+PV+PVG	V+G+VG+PG+PV+PVG	V+G+VG+PG+PVG	V+G+VG+PG+PV+PVG	gage = gate 1

The setting of the override selector in conjunction with the "VIS" and "GAGE" selector switches determine which reject pile receives sheets with the various defect combinations. This chart shows the possibilities available. Note that any one of seven defects or defect combinations can occur in a sheet. These are designated P, V, G, PV, PG, VG, PVG.

"open 1" sensor before the second sheet is sheared. In this case, the first leading edge to activate the "open 1" sensor following the shear cut will be the sheet being sheared and proper operation will result.

On the other hand, if the sheet spacing is too close, it may be that the first leading edge to operate the "open 1" sensor will be the free sheet rather than the sheet being sheared. In this case, the wrong sheets will be rejected. If this does occur, it will be necessary to readjust the open 1 sensor. At the same time, it may also be necessary to readjust the length of the conveyor shift register since the distance between the open 1 and open 2 sensors would increase.

Optional manual gate operation can be obtained by jumpering terms 816 to 817 (gate 1) and 807 to 817A (gate 2). NOTE: This function is not synchronized with sheet motion; therefore, the gates will operate whenever the circuits are jumpered. This function should not be used with the line running unless the customer uses external synchronizing and interlocking circuits.

MAINTENANCE

This equipment requires a minimum amount of maintenance. The maintenance required is preventative in nature and it can eliminate problem spots before they appear.

The lamps in the pulse tachometers should be replaced on a regular basis - do not wait for a failure. Lamps age, and even though they still produce light, their output can be greatly reduced by blackening on the inside surface of the glass bulb. This type of situation can cause intermittent problems which may be difficult to diagnose. These problems can be prevented by keeping spare lamps on hand and replacing the lamps every 5,000 hours.

SERVICING

Effective servicing of this equipment can best be done by personnel who have a thorough understanding of the classifier circuits and the logic elements it uses. In order to gain this knowledge, the serviceman should study this instruction book and the diagrams pertaining to the equipment.

It is essential that proper test equipment be used in servicing. In most cases, static logic levels can be measured with a conventional multimeter. This meter should have a sensitivity of 20,000 ohms/volts. Dynamic circuit tests can best be made with an oscilloscope.

The test panel, which is built into this equipment, is a most useful servicing aid. The serviceman should become familiar with the switch functions and the meanings of the various indicator lights so that he can put the test panel to effective use.

An extender board, supplied with the equipment, is also useful in servicing. A board which may be in question is plugged into the extender and the extender is then plugged into the rack. This makes the circuit points, on the questionable board, accessible for testing.

One of the best aids in servicing and also a deterrent to prolonged shutdowns is a set of spare parts. Many of the classifier functions are on plug-in units and it is recommended that a set of these units be kept on hand. With these units, whole circuits can be quickly exchanged to get the line back into operation. Then, the defective unit can be repaired at a more convenient time.

Note that care in servicing this equipment is important. Accidental short circuits, even though momentary, can damage circuit components.

TROUBLESHOOTING

This information is intended as an aid in localizing a trouble to a particular portion of the circuit or to a particular sub-panel. Once a problem is localized to a replaceable sub-unit, it is recommended that the whole panel or device be replaced from a spare parts inventory. In this manner, the equipment and shear line can be restored to operating condition quickly, thereby minimizing shutdown time. Later, when the line is purposely shut down (overnight or for the weekend), or in the repair shop, the faulty sub-unit can be repaired without affecting the line output.

POWER SUPPLIES

A check of the power supplies is usually a good first step in finding a trouble. Check the 115V AC input and then the internal supplies. If a fuse is blown or an internal supply is dead, check for short circuits or excessive loads before installing a new unit and reapplying power.

NOTE: Do not perform operation tests on battery power alone. Line voltage must be on.

INPUT CIRCUITS

Set the "run-test" switch to "run" and simulate an input from the device in question (for example:

energize sheet sensors with a piece of steel, push visual reject pushbuttons, generate a pin hole signal with pin hole test buttons - etc.). The appropriate indicator should light. If it does not, check for an input signal at the input of the appropriate isolator or relay. If the proper signal is being received, check at the isolator or relay output. Replace the isolator if no signal is present. If the signal is present at the isolator output, replace the plug-in board which contains the input circuit in question. Use input and output voltage checks at the pins in question to verify the problem area. Be sure to check to make sure that the input jumpers on the 3S7517SR300 boards are set properly. An improper setting could appear to be a missing input signal.

TEST CIRCUITS

There is the possibility of a problem in the test circuits. A trouble here can be quite confusing since the line may function normally while the test panel indicates abnormal operation. A few input and output voltage checks should pinpoint the problem to a defective indicator LED (light emitting diode) or the driver circuit. If this is the case, replace the LED or board containing the defective driver circuit.

SHIFT REGISTERS

A signal, once inserted into the register, should flow through (in auto shift) and after the proper number of shifts (time delay), appear at the appropriate accumulator. The input and output and in most cases some intermediate points in the register are indicated on the test panel. If the signal is lost in a register, replace the 3S7517SR300 in which it is lost. Since there are several 3S7517SR300 boards in the control, it is possible to find or verify a problem board by swapping; however, always note the jumper settings - improper settings will cause improper operation. Note that if a jumper screw does not make contact (loose or dirty), continuous defect signals will result - check and tighten all jumpering screws if this symptom is found.

If a register functions in the "test" mode but not in "run", check for proper tachometer signals and associated circuits. Use the "manual shift" to check the strip tachometer divider - 12 pulses in yield 1 pulse out.

OVERRIDE CIRCUITS

The override circuits are located on the 3S7517SS250 board. Check for proper control inputs from the selector switch. Using the chart in the initial adjustments section, set up the third priority signal first (use test panel switches and auto shift to set appropriate accumulator, P, V, or G). Then insert the second priority signal - the other should reset. Finally, insert the #1 priority signal - the other should reset. This procedure can be used for each override selector switch setting. If a problem is found, replace the 3S7517SS250 board or use input and output voltage checks to isolate the defective component.

OUTPUT CIRCUITS

With the line shut down, check for gate operation by jumpering the manual gate control inputs at 2TB; for gate 1, 816 to 817 and 807 to 817A for gate 2. If the gate in question does not operate, check for proper voltages and/or the external circuits; if it does, check the static switches. If the proper signals are not being applied to the static switch(es), change or check the 3S7517SH250 board.

RENEWAL PARTS

When ordering renewal parts, address the nearest General Electric Sales Office, specify the quantity required, and give the catalog numbers or describe the required parts in detail. In addition, give the 3S number and a complete description of the equipment for which the part is required.

PRINCIPAL RENEWAL PARTS LIST

QTY.	PART NUMBER	DESCRIPTION
1	3S7517SH250A2	Plug-in Board
1	3S7517SS250A2	Plug-in Board
1	3S7517ST250A2	Plug-in Board
1	3S7517TP250A2	Plug-in Board
4	3S7517SR300A1	Plug-in Board
1	3S7517EA300A1	Plug-in Input Isolator (Voltage Pulse)
6	3S7517EA304A1	Plug-in Input Isolator (Contact Input)
1	3S7517EA305A1	Plug-in Input Isolator (Dual Tach Sigs)
1	CR120HG85C02	Relay (LVR)
5	44A339874-001	Relay (G1, G2, VIS, LOK and SVR)
3	GSR10AU5	Static Switch (1SS, 2SS and MSS)
10	44A370813-008	Switch SPST Spring Return (2 through 5SW, 7 through 10SW, 16 and 17SW)
6	44A370813-006	Switch SPST (1SW, 6SW, 12 through 15SW)
1	44B214628-001	Switch (11SW)
23	44A335848-001	LED (1-23 LD)
1	3JX5008	Line Filter (Sprague)
1	K9774741P4	Fuse .1 amp (3FU)
2	K9774700P4	Fuse 10 amp (1 and 2FU)
1	44A336717-001	Battery (Globe GC 426-1)
2	44A335813-001	5V DC 5A Power Supply
3	GE5042-2	Switch, DPST (LSW, 1GSW and 2GSW)
4	44A334272-001	MOV, 115V (1 through 4 MOV)
2	1N93	Diode (2 and 3D)
6	44B232011-001	Diode (1, 4 through 8D)

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